

1. ① 至少需要  $\lceil \log_2 51 \rceil = 6$  位二进制

② 二进制: 00110011

十六进制: 33

BCD码: 0101 0001

③ 11100011

④ 51: 0101 0001

+ 29: 0010 1001

$$\begin{array}{r} 0101\ 0001 \\ + 0010\ 1001 \\ \hline 0111\ 1010 \\ + \quad \quad 0110 \\ \hline 1000\ 0000 : 80 \end{array}$$

3. 证: 1)  $X + 1 = X + (X + X')$  (互补律)

$$= (X + X) + X' \quad (\text{结合律})$$

$$= (X + X) \cdot 1 + X' \quad (\text{恒等性})$$

$$= (X + X) \cdot (X + X') + X' \quad (\text{互补律})$$

$$= X + (X \cdot X') + X' \quad (\text{分配律})$$

$$= X + 0 + X' \quad (\text{互补律})$$

$$= X + X' + 0 \quad (\text{交换律})$$

$$= 1 + 0 \quad (\text{互补律})$$

$$= 1 \quad (\text{恒等性})$$

2)  $X + X'Y = X + X'Y$  (恒等性)

$$= X \cdot (1 + Y) \quad (\text{分配律})$$

$$= X \cdot 1 \quad (\text{空单元})$$

$$= X \quad (\text{恒等性})$$

4.

|   | A | B | C | D | f |
|---|---|---|---|---|---|
|   | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 1 | 0 |
|   | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 | 0 |
| 2 | 0 | 1 | 1 | 1 | 0 |
| 3 | 0 | 1 | 0 | 1 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 |
| 5 | 1 | 1 | 0 | 0 | 1 |
| 6 | 1 | 1 | 0 | 1 | 1 |
| 7 | 1 | 1 | 1 | 1 | 1 |
| 8 | 1 | 1 | 1 | 0 | 1 |
| 9 | 1 | 0 | 1 | 0 | 1 |
|   | 1 | 0 | 1 | 1 | 0 |
|   | 1 | 0 | 0 | 1 | 0 |
|   | 1 | 0 | 0 | 0 | 0 |